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CT imaging features of canine thymomas

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Abstract

Canine thymomas have been evaluated based on clinical features, treatment options, surgical excision, and outcomes with limited information on specific CT features. The objective of this retrospective, descriptive, cross-sectional study was to describe the CT characteristics of confirmed thymomas and to compare these imaging features to outcome. A total of 22 dogs met the inclusion criteria of histologically confirmed thymomas with concurrent CT imaging. Tumor size varied widely ranging from small and well-circumscribed to large and invasive. Delayed-phase, contrast-enhanced CT studies were best for determining the degree of contrast enhancement in tumors. Of these, 19 of 22 masses had heterogeneous enhancement and three of 22 masses had homogeneous enhancement. Vascular invasion was present in seven of 22 cases. Larger tumors were associated with vascular invasion (height: $P = .04$; width and volume: $P = .02$). On precontrast CT, larger tumors (16/21) were heterogeneous and cystic, with smaller tumors (5/21) being more homogeneous (all values $P < .05$). A larger size was associated with recurrence in fully resected masses (height: $P = .03$), but not a shorter outcome ($P > .3$ for all size dimensions). Postoperative complications and incomplete tumor resection were associated with shorter outcome (both values $P < .01$). Metastasis was confirmed in four cases. There were six cases with lymphadenopathy noted on CT; five of the six cases did not have evidence of metastasis. Larger tumors were more likely to be cystic and associated with vascular invasion.

KEYWORDS

angiogram, cranial vena cava, mediastinal mass, neoplasm

1 | INTRODUCTION

Thymomas, the second most common mediastinal mass in dogs, are neoplasms originating from the epithelial portion of the thymus, characterized primarily by neoplastic epithelial cells but containing variable amounts of non-neoplastic lymphocytes and occasionally mast cells and eosinophils.[1–6] Thymomas are characterized by their benign cytologic and histologic features but may display clinical behavior ranging from benign to malignant with aggressive features including extension into surrounding vasculature with or without metastasis.[2,4,5] Canine thymomas typically affect middle-aged to older dogs and have been reported to be more common in Labrador Retrievers and German Shepherds.[1,4,7] Common clinical presenting signs may include vomiting, regurgitation, anorexia, cough, dyspnea, tachypnea, and lethargy.[4,5,7–10] Thymomas are commonly associated with paraneoplastic syndromes, including myasthenia gravis, cranial vena cava syndrome, and hypercalcemia of malignancy.[4,5,10–13]

Thymomas have been described radiographically as a soft-tissue opacity in the ventral cranial mediastinum that may cause displacement of the heart and trachea.[5,14] Megaesophagus and an alveolar infiltrate in the lung parenchyma due to aspiration pneumonia can be associated with thymomas.[1,3,4,12,14,15] One study evaluating the ultrasonographic appearance of thymomas reported that thymomas are more likely to have a heterogeneous echogenicity and one or more internal cysts when compared to lymphoma, which is more uniform in echogenicity.[3] Recently, contrast-enhanced ultrasound of thymomas showed different enhancement patterns of thymomas compared to lymphoma.[16] Computed tomography imaging has been used to evaluate thymomas with imaging findings of a heterogeneously contrast-enhancing, soft-tissue attenuating, cranial mediastinal mass with or without central cystic cavitation.[8,9] Adjunct findings may include lymphadenopathy, displacement or invasion of surrounding vessels, and in rare cases presence of pleural effusion or metastasis.[1,8–10,12,14,17]

The purpose of this study was to further characterize CT features of thymomas. Additional aims included evaluation of specific imaging features and patient outcome to determine if any specific imaging features correlate to poorer outcome. Our hypothesis was that larger

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tumors and those with vascular invasion would be more likely to contribute to a poorer outcome.

2 | METHODS

The study was a retrospective, descriptive, cross-sectional design. Medical records of dogs presented to the Colorado State University Veterinary Teaching Hospital between 2008 and 2018 that had a CT scan of the thorax were evaluated for inclusion in the study using the terms “thymoma” and “mediastinal mass.” The hospital director approved the use of patient data for this study. Criteria for inclusion were a CT scan of the thorax and histopathologic confirmation of thymoma. All decisions for subject inclusion or exclusion were made by a clinical veterinarian (L.v.S.).

Information collected from the medical records included signalment, presenting complaints, clinical history, histopathology reports, as well as date and cause of death when available. Medical record data were recorded by a clinical veterinarian (L.v.S.), who was not aware of CT findings at the time of data recording.

2.1 | Image evaluation

All CT images were reviewed by two American College of Veterinary Radiology-certified veterinary radiologists (A.M., E.R.) to reach a consensus. These observers were blinded to medical record findings at the time of CT image evaluation. Computed tomography images were reviewed on a digital workstation (Philips IntelliSpace PACS Radiology). The following data were recorded for each CT study: mass laterality (left, right, central), mass size, mass volume, attenuation of the mass, type of contrast enhancement (postcontrast or three-phase angiography), and presence of: vascular invasion, pulmonary nodules, or lymph node enlargement. Esophageal dilation was not assessed due to concerns that general anesthesia could cause esophageal dilation unrelated to the thymoma. Mass laterality was defined according to the location of over 50% of the mass; when the mass location was equal on each side of midline this was defined as central. To determine size of the mass at the largest dimensions, length measurements were performed on the dorsal plane reconstructed view, width measurements on the transverse plane view, and height measurements on the sagittal plane reconstructed view. Tumor volume was obtained using tumor tracing techniques through proprietary software (Philips IntelliSpace Portal, Amsterdam, the Netherlands) similar to previous literature on tumors of irregular contour (Figure 1).[18] Attenuation of the mass was defined as either uniformly soft tissue attenuating, with Hounsfield units (HU) between 35 and 65, or heterogeneous to cystic, with areas of HU < 25 in addition to areas of HU between 35 and 65. Contrast enhancement of the mass was evaluated in arterial, venous, and delayed phases of the scan with enhancement being defined as over 15 HU change compared to the precontrast portion of the study. Categories of contrast enhancement included homogeneous, heterogeneous, peripherally enhancing, or minimal to no enhancement. Technical information about the available CT scans was also recorded.

2.2 | Outcome data

Outcome data were recorded by a clinical veterinarian (L.v.S.) who was unaware of CT findings. Outcome was defined as death resulting from the surgical procedure or progressive disease. Patients were censored if they were lost to follow-up or if death was due to another cause. Time to outcome was categorized into four categories, with 0: survival < 1 month; 1: survival of 1–6 months; 2: survival of 7–12 months; 3: survival of >1 year. Variables compared to outcome included the following: tumor size (height, width, length, and volume), presence of aspiration pneumonia, myasthenia gravis, presence of vascular invasion on CT, tumor resectability, and postoperative complications. The presence of aspiration pneumonia, myasthenia gravis, tumor resectability, and postoperative complications were also recorded from medical record entries.

2.3 | Statistical analysis

Statistical analyses were performed by a university employed statistician (S.R.), using statistical analysis software (SAS v9.4, SAS Institute Inc., Cary, NC). Descriptive continuous data was evaluated using means and SDs. Normality was determined prior to performing *t*-test to compare two groups and ANOVA to compare more than two groups. If normality was not met, the data was represented using medians and analyzed using a nonparametric Wilcoxon test or Kruskal-Wallis test. The categorical data was represented using percentages and a chi-square test was used to compare. If any of the categories had less than five numbers, Fisher’s exact test was used. A *P*-value of .05 was used to determine statistical significance.

3 | RESULTS

3.1 | Demographics and clinical history

A total of 22 dogs met the inclusion criteria. The mean age of the dogs was 10.4 years with a median of 10 years (range: 6 to 15 years). Labrador Retrievers accounted for eight of the 22 cases (36%), followed by mixed breed (n = 4), Golden Retriever (n = 2), and Border Collie (n = 2) with one each of several other breeds. There were nine females and 13 males. All animals were spayed or neutered aside from one intact male. The most common presenting clinical signs included coughing (40.9%), regurgitation (31.8%), dyspnea, weight loss, and inappetence (all 22.7%).

Four dogs (18.1%) had myasthenia gravis, three of which were confirmed with acetylcholine receptor antibody titers, and one of which was confirmed with a tensilon test. A fifth dog was tested for myasthenia gravis, which was negative, and the remainders of the dogs (17/22) were not tested and thus their status was unknown. Although four dogs’ tumors were nonresectable, debulking surgery was performed. Masses were nonresectable in these cases due to either firm adherence to the great vessels or to mediastinal structures such as the pericardium. Vascular invasion itself was not a determinant

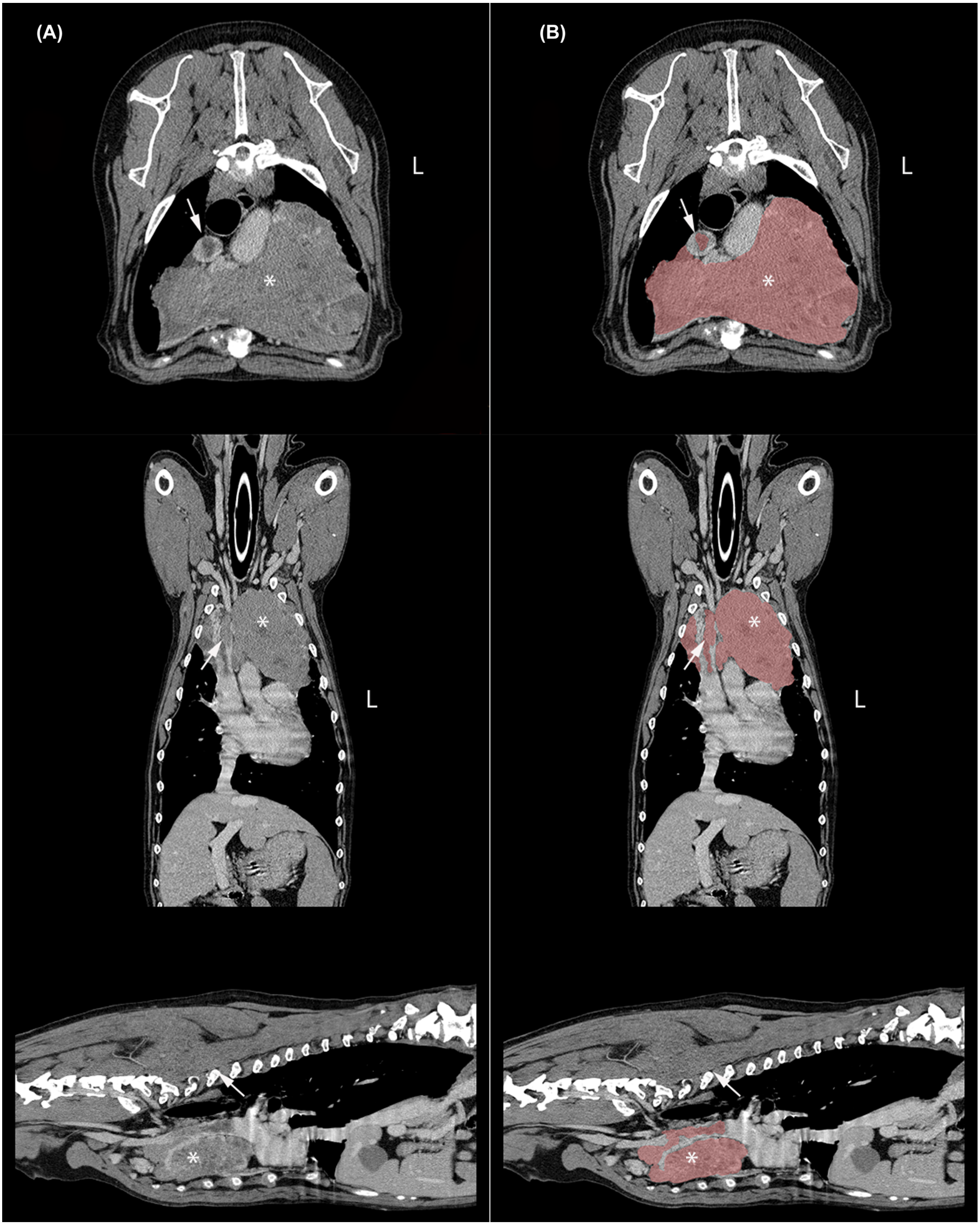

FIGURE 1 Transverse, dorsal, and sagittal CT images showing use of volume tracking software (pink colored regions) to estimate volume of a large thymoma (*) with vascular invasion (arrows) within the venous phase of the angiogram in a standard algorithm (Slice thickness = 2 mm, Window = 400, Level = 60). This mass was categorized as left-sided due to greater than 50% being on the left side of the thorax. A, without volumetric tracking. B, with volumetric tracking

of resectability as four tumors with vascular invasion were able to be fully resected at the time of surgery. Five of the 22 dogs (22.7%) had postoperative complications including pyothorax (n = 2), overall poor recovery from surgery (n = 1), and complications likely secondary to a history of myasthenia gravis, including increased regurgitation, weakness, and dyspnea (n = 1) and aspiration pneumonia (n = 1). Five dogs (22.7%) overall were euthanized within a month of surgery including four of the dogs mentioned that had postoperative complications as well as a fifth dog with a nonresectable tumor that had progression of its invasive mass into the cranial vena cava and right atrium. One dog received radiation therapy on the day of surgery as well as a second dose 7 days postoperatively and a second dog received radiation therapy 9 months postoperatively when there was recurrence of the mass. The first dog was euthanized one week later due to a pyothorax and the second dog was euthanized 14 months postoperatively due to declining quality of life associated with osteoarthritis.

TABLE 1 CT imaging features of thymomas in 22 dogs

CT characteristic category	Measured values
Laterality	
Left	10/22 (45.5%)
Central	9/22 (40.9%)
Right	3/22 (13.6%)
Enlarged lymph nodes	5/22 (22.7%)
Presence of vascular invasion	7/22 (31.8%)
Attenuation pre-contrast	
Soft tissue attenuating	5/21 (23.8%)
Heterogeneous/cystic	16/21 (76.2%)
Size	
Height (cm)	2.57-11.82 [mean, 6.4]
Width (cm)	2.47-17.61 [mean, 8.0]
Length (cm)	2.76-21.83 [mean, 10.6]
Volume (cm3)	10.77-1875.51 [mean, 459.6]

3.2 | Imaging results

Twenty of the CT studies, including all angiograms, were performed on a 16-slice CT scanner (Gemini TF Big Bore System, Philips Medical System, Cleveland, OH, USA). The remaining two CT scans were performed on single slice CT scanners (Picker PQ 2000, Picker International, Highland Heights, OH, USA; Toshiba Aquilion, Toshiba America Medical Systems, Duluth, GA). Helical images with a slice thickness of 1 to 5 mm and pitch of 0.94-1.6 were acquired of the thorax. The tube rotation was 0.5-1 s with a current ranging from 125 to 469 mA and a voltage of 120 to 130 kVp. The field of view ranged from 187 to 450 mm and matrix dimensions were either 512 × 512, 768 × 768, or 1024 × 1024. For contrast studies, iohexol (Omnipaque; GE Healthcare Inc, Princeton, NJ, USA) was administered intravenously in the cephalic vein at a dose of 2 mL/kg of body weight (concentration of 350 mg/mL). A power injector (Stellant Dual Head Power Injector; Medrad, Indianola, PA, USA) was used for all angiogram studies with a rate of 3 mL/s for all patients under 15 kg and 4 mL/s for patients over 15 kg. Computer software determined timing of arterial and venous phases based on time of contrast appearing in the aorta. Delayed phase contrast images were acquired approximately 2-3 min post iodinated contrast medium injection. A soft tissue reconstruction algorithm was available for most studies with a window width of 400 and window level of 40 to 60. For several studies, a lung reconstruction algorithm was also available, with a window level of –600 and width of 1600.

When evaluating laterality on CT, 10 of 22 (45.45%) were left-sided, nine of 22 (40.91%) were central, and three of 22 (13.64%) were right-sided (Table 1). Tumor size varied widely from 2.96 cm (H) × 2.69 cm (W) × 2.76 cm (L), to 11.82 cm (H) × 17.61 cm (W) × 21.83 cm (L). Tumor volume also varied widely, from 10.77 to 1875.51 cm^3. Fourteen of 22 CT studies had three phase angiograms performed. Seven of eight of the remaining cases had both pre- and postcontrast examinations available, with one case only having a post-contrast examination available at the time of evaluation.

Precontrast attenuation was evaluated in 21 of the 22 cases. Sixteen out of 21 cases (76.2%) had heterogeneous or cystic attenuation, and five of 21 (23.8%) cases were uniformly soft tissue attenuating. Of the 14 cases with angiograms available, 10 of 14 (71.4%) were heterogeneously contrast-enhancing in the arterial phase, while the remaining four of 14 (28.6%) had no visible contrast enhancement in the arterial phase (Table 2). In the venous phase, all 14 of 14 (100%) of the tumors were heterogeneously contrast-enhancing. No tumors were homogeneously or peripherally contrast-enhancing in the arterial or venous phases. In the delayed contrast phase, 19 of 22 (86.4%) were heterogeneously contrast-enhancing, while 3/22 (13.6%) were homogeneously contrast-enhancing (Figure 2).

Vascular invasion was noted in seven of 22 cases (31.8%), with all seven cases invading the cranial vena cava (Figure 3). In one case, vascular invasion extended from the cranial vena cava into the right atrium, and in a different case vascular invasion was also present within both brachiocephalic veins and the left costocervical and internal thoracic veins. Pulmonary nodules were noted in three cases. In two patients, these nodules were not neoplastic based on necropsy. In the third patient, these nodules were of unconfirmed origin. Lymphadenopathy was noted in six cases (27.2%), including sternal (n = 5) lymphadenopathy and cranial mediastinal (n = 1) lymphadenopathy. There were four confirmed cases of metastasis through surgical biopsies with two metastatic sternal lymph nodes, one metastatic cranial mediastinal lymph node and a metastatic thoracic wall mass. Only one of these cases had lymphadenopathy on CT that correlated with the histopathologic confirmation of metastasis. Mild pleural effusion was noted in two cases.

3.3 | Size comparison to other imaging features

When comparing height of left-sided tumors (mean = 5.1, median = 4.86 [3.05-8.11]) to centrally located tumors (mean = 7.86,

TABLE 2 CT contrast enhancement characteristics in 22 dogs with thymoma

Contrast enhancement category	Arterial phase	Venous phase	Delayed phase
Homogeneous	0/14 (0%)	0/14 (0%)	3/22 (13.6%)
Heterogeneous	10/14 (71.4%)	14/14 (100%)	19/22 (86.4%)
Peripherally contrast enhancing	0/14 (0%)	0/14 (0%)	0/22 (0%)
Minimal/None	4/14 (28.6%)	0/14 (0%)	0/22 (0%)

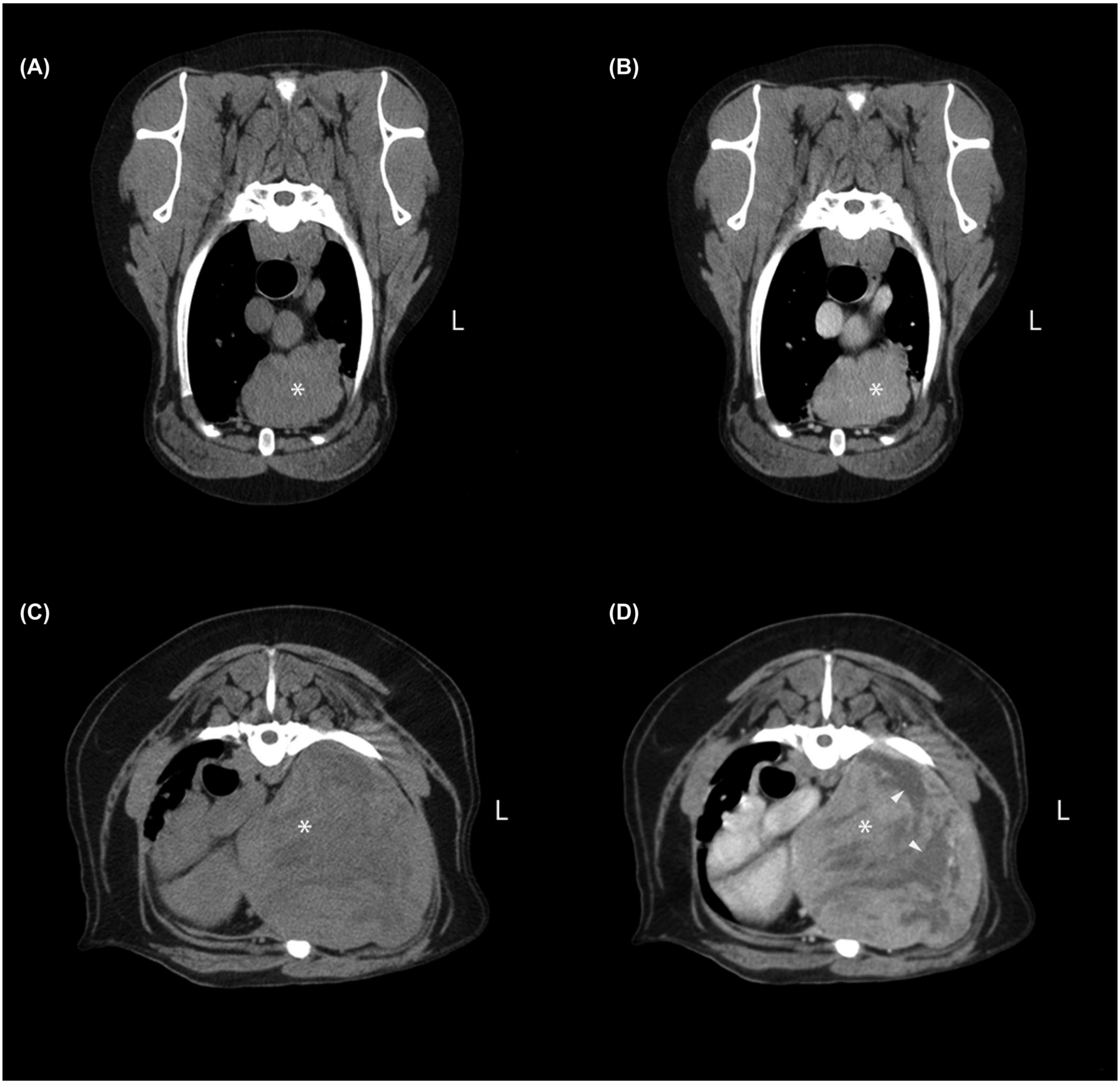

FIGURE 2 Transverse CT images of both a homogeneously (A, B) and heterogeneously (C, D) contrast-enhancing thymoma (*) in a standard algorithm, 3-min delayed phase of the angiogram. (Slice thickness = 2 mm, Window = 400, Level = 40). Note the large noncontrast enhancing, cystic components (white arrow heads) in the heterogeneously contrast-enhancing mass (D). A and C, Precontrast. B and D, Postcontrast

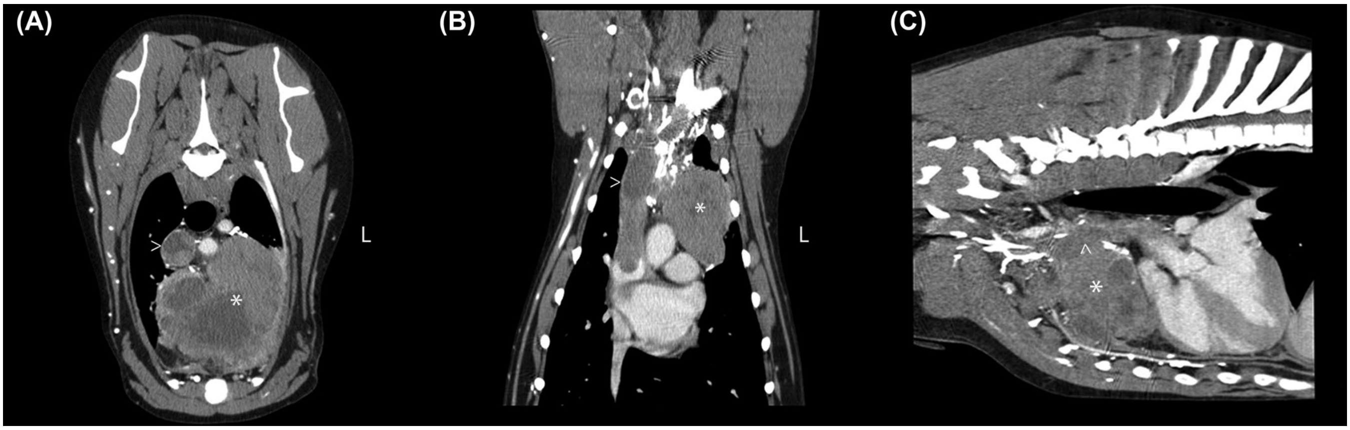

FIGURE 3 Postcontrast CT images acquired in standard algorithm in the venous phase of the angiogram. Note the large heterogeneously contrast-enhancing thymoma (*) in the cranial mediastinum with invasion into the cranial vena cava (arrowheads). (Slice thickness = 2 mm, window = 400, level = 40, algorithm = soft tissue). A, Transverse plane. B, Dorsal plane. C, Sagittal plane

median = 9.45 [2.57-11.82]), larger tumors were more centrally located, although this finding was not significant (P = .08). Similarly, when comparing width of left-sided tumors (mean = 5.36, median = 3.87 [2.47-15.05]) to centrally located tumors (mean = 10.57, median = 11.27 [2.69-17.61]), larger tumors were more centrally located, although this finding was also not significant (P = .06). Additionally, no significance was found in evaluating length of tumors (P = .10) or overall volume (P = .12) with laterality, and right-sided tumors were excluded from this analysis due to small sample size.

Precontrast attenuation was significantly associated with all size parameters as well as volume (all values < 0.05), with smaller tumors being more uniformly attenuating and larger tumors more heterogeneous and cystic (Figure 4). Tumor heterogeneity based on contrast enhancement was not significantly associated with tumor size or volume (all P-values > .35). In four studies, the arterial phase differed from the delayed phase. All four cases exhibited no contrast enhancement in the arterial phase. In the delayed phase, three cases exhibited heterogeneous contrast enhancement and one case exhibited homogeneous contrast enhancement.

Vascular invasion was significantly associated with larger tumor size in multiple planes (height, P = .04; width, P = .02) and with larger tumor volume (P = .02; Figure 5). No significance was found in evaluating vascular invasion compared to laterality or to presence of lymphadenopathy. A clinical history of regurgitation was not found to have significant association with either laterality, size in any dimension, or volume of masses evaluated on CT.

Lymph node enlargement was significantly associated with tumor width (P = .03), with tumors with a greater width more likely to have lymphadenopathy. However, lymph node enlargement was not significantly associated with height (P = .25), length (P = .08), or volume (p = 0.11). None of the five homogeneously attenuating masses had evidence of lymph node enlargement, whereas six of 16 (37.5%) of the heterogeneously attenuating masses showed lymph node enlargement.

3.4 | Outcome data

Thymomas with larger height measurements were found to be significantly associated with recurrence (P = .03), although neither length (P = .08), width (P = .08), nor volume (P = .10) measurements were significantly associated with recurrence. However, no size parameters were associated directly with a poorer outcome. Megaesophagus and myasthenia gravis were not found to be significantly associated with a shorter outcome (P = .28). The presence of aspiration pneumonia was closely associated with outcome, but this was not significant (P = .06). Vascular invasion was not significantly associated with a shorter outcome (P = .08). Finally, both the presence of nonresectable tumors (P < .01) and the postoperative complications were found to be significantly associated with a shorter outcome (P < .01).

4 | DISCUSSION

Findings from the current study indicated that CT evaluation allowed for detailed characterization of tumor size and that larger thymomas were more likely to have vascular invasion. Smaller tumors were more likely uniformly attenuating, while larger tumors were heterogeneous and cystic. Heterogeneous contrast enhancement was not associated with caval invasion. Contrary to our hypothesis, neither larger volume nor vascular invasions were associated with a poorer outcome. Further information from CT images can include identifying volume occupied by nonuniform masses using novel tumor tracking techniques.

Size varied widely in our study, with the majority of the tumors being large, space-occupying masses in the thoracic cavity. This is likely due to disease diagnosis at later stages and the start of clinical signs such as coughing, tachypnea, or dyspnea. Three of the smaller tumors within the study were discovered incidentally in completely asymptomatic animals. In this study, larger tumors were associated with vascular

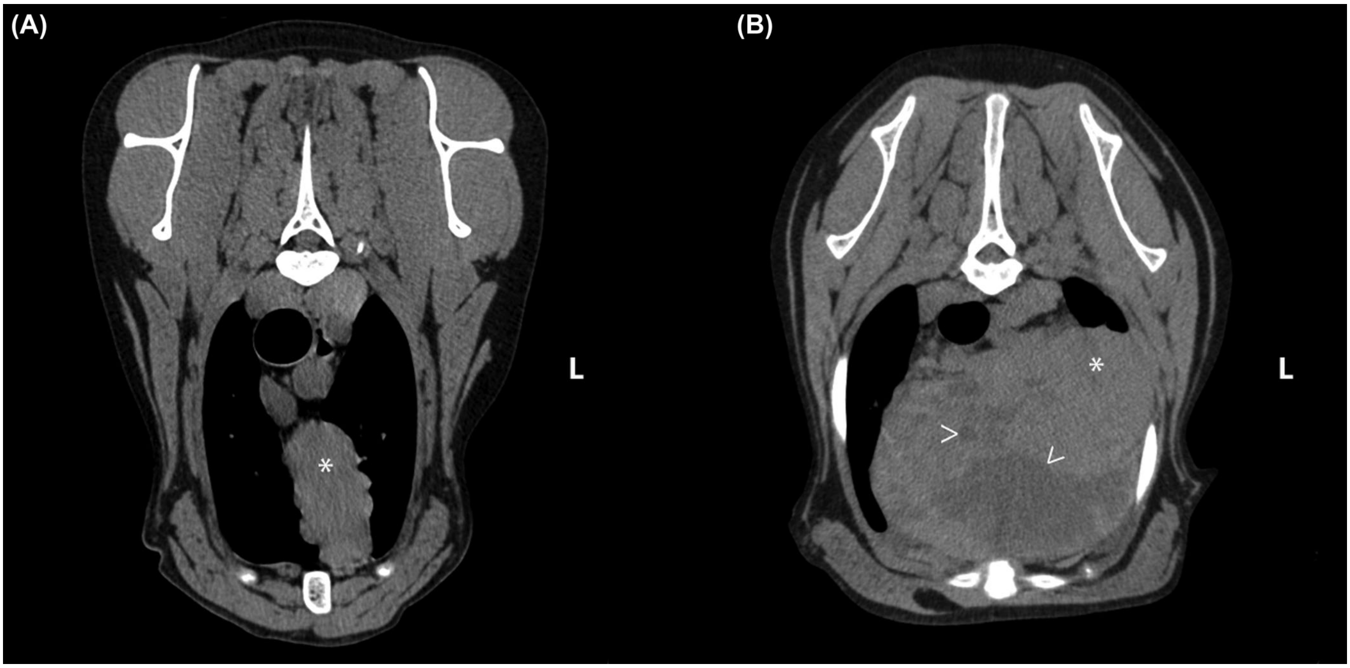

FIGURE 4 Transverse precontrast CT images in two dogs with thymomas. A, There is a small thymoma (*) that is uniformly soft tissue attenuating. (Slice thickness = 1 mm, Window = 400, Level = 60, algorithm = soft tissue). B, This is an example of a larger thymoma (*) that is heterogeneous, with multiple internal cystic regions (arrowhead). (Slice thickness = 2 mm, window = 400, level = 40, algorithm = soft tissue)

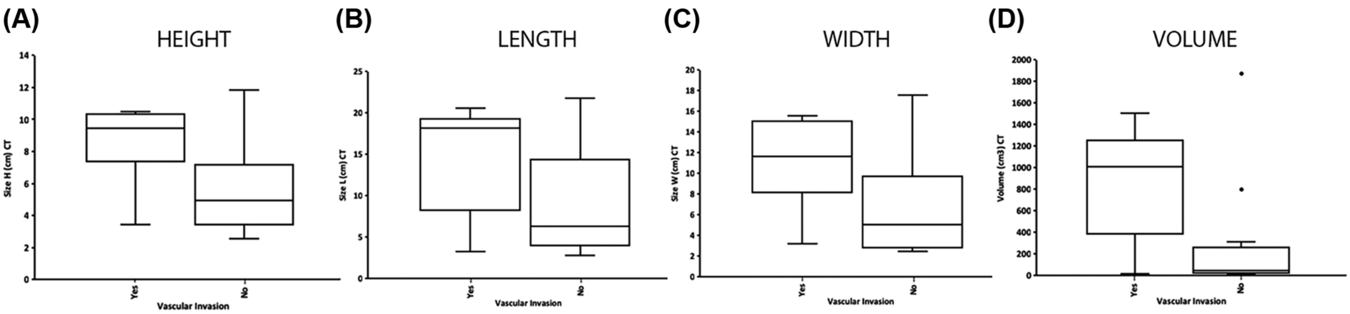

FIGURE 5 Box and whisker plots comparing the presence of vascular invasion and its association with thymoma size (A, height; B, length; C, width) as well as volume (D). There was a significant difference with height ($P = .0409$), width ($P = .0241$), and volume ($P = .0289$) of thymomas and vascular invasion, with larger thymomas being positively associated with vascular invasion. Length was not significantly different ($P = .0907$)

invasion; however, several smaller tumors showed aggressive behavior and vascular invasion. Neither a larger size, volume, nor vascular invasion was found to be associated with outcome.

As expected, improved internal detail was appreciated on CT with the majority of cases being heterogeneous or cystic. This supports a previous ultrasound study, in which 57.14% of thymomas were cystic.[3] This is also consistent with previous case reports of canine thymomas, in which they have been noted to be heterogeneous and cystic on CT.[8,9,19] This study confirms the cystic and heterogeneous attenuation of thymomas precontrast and also has shown a correlation between size and development of internal cysts. Larger thymomas were much more likely to be internally cystic. This is in contrast to human thymomas, where a previous retrospective study showed 18.75% had a cystic component.[20] It is postulated that this heterogeneity of canine thymomas could help to discriminate from mediastinal lymphoma, as only three of 15 cases of lymphoma in the ultrasound study were cystic.[3]

Most thymomas had more pronounced contrast enhancement in the venous or delayed phases. Tumors that appeared homogeneous in the arterial phase were more heterogeneous in the venous or delayed phases, suggesting additional perfusion of the tissues in the venous or delayed phases or early washout of some areas from the arterial phase. In a recent paper evaluating contrast-enhanced ultrasonographic characteristics of intrathoracic mass lesions, three of seven (43%) thymomas showed a heterogeneous and cystic parenchyma, four of seven (57%) thymomas showed heterogeneous enhancement, while eight of eight (100%) of lymphomas enhanced uniformly.[16] In humans, heterogeneous enhancement on CT was suggestive of an invasive thymoma or thymic carcinoma.[20] This is in contrast to our study, where most of the masses had heterogeneous enhancement, but only a small subset were invasive thymomas with caval invasion. Previous case reports of canine thymomas have also involved heterogeneous masses without presence of caval invasion.[9,19]

Laterality on CT imaging, or the presence of an off-midline mass, has been found in human studies to favor a diagnosis of thymoma over that of lymphoma.[21] On CT evaluation of all 22 dogs, 45.45%(10) were left-sided, 40.91%(9) were centrally located, and 13.64%(3) were right-sided. As the thymus is a left-sided organ in younger animals, an explanation for the large percentage of centrally located tumors is likely attributed to size. When evaluating size with laterality, as height and width increased, thymomas were more centrally located likely because they occupy much of the cranial thoracic cavity as they grow. Further studies would be required in order to evaluate if there is a significant difference in laterality between canine thymomas and mediastinal lymphoma.

Six cases of lymphadenopathy were noted on CT. Computed tomography provides sensitive evaluation of lymph node size, shape, and attenuation, and previous studies have correlated larger size to the presence of metastasis.[22] In our study, metastasis was confirmed on histopathology in one case of sternal lymphadenopathy, but five other cases of lymphadenopathy did not reveal evidence of metastasis. Two of the cases in which there was metastasis present on sternal lymph node biopsy there was no lymphadenopathy noted, suggesting that sternal lymph node biopsies should be considered for more accurate staging. As sternal lymph nodes receive drainage from the thymus through their afferent vasculature, this may be the lymph node to preferentially remove in surgery for histopathologic evaluation for metastasis.[22]

Treatment for thymomas is primarily surgical excision and all dogs included in this study had surgery for full resection or debulking.[14] Postoperative complications and inability for full tumor resection were significantly associated with outcome. One previous study comparing results of excision of thymomas in cats and dogs found that animals with invasive or nonresectable tumors had a higher mortality rate but had good long-term survival if they survived the first 24 h.[14] In humans, the ability to fully resect a tumor has been shown to be the most important prognostic factor.[20] In our study, the inability to completely remove the full size of the tumor was associated with a poorer outcome.

Two dogs in our study received radiation therapy at different points following surgery. In a study, evaluating a standardized protocol of hypofractionated radiation therapy in dogs with thymomas not undergoing surgery found that radiation therapy was well-tolerated and produced a response in four of eight (50%) of the patients.[10] Two other patients with stable disease still achieved long-term survival of over 2 years.[10] The dogs in our study received far fewer radiation treatments as well as total doses and radiation therapy was not thought to contribute to their outcome. Further studies will be needed to fully evaluate the benefit of radiation therapy in dogs with thymomas, particularly in cases of non-resectable disease.

Limitations of this study include its retrospective nature and small sample size. Certain data were not available in all cases due to the retrospective nature including radiographs, major histopathologic subtype, and complete diagnostic testing for myasthenia gravis. This may have led to underrepresentation of myasthenia gravis in this study. The CT scans were acquired on multiple different scanners, which may have affected imaging characteristics, especially density values. Additionally, for certain cases, the omission of CT angiography may have hindered evaluation of the vasculature and invasion. Also, inclusion criteria bias may have played a role in case selection as all dogs required histological diagnosis of thymomas. Lastly, the radiologists reviewing the imaging studies were aware that all dogs had confirmed thymomas but were blinded to history and outcome information. Future prospective studies utilizing CT angiography may be beneficial for further evaluation of canine thymomas.

In conclusion, larger thymomas are more likely to be cystic in nature and have concurrent vascular invasion. Given this finding, identification of larger thymomas on radiographs may warrant CT evaluation for vascular invasion prior to surgery. Both postoperative complications and incomplete resection were associated with poorer outcome (i.e., earlier death). Finally, lymphadenopathy noted on CT was not a useful predictor of metastasis. Further studies are needed to better determine factors affecting outcome in dogs with thymomas.

LIST OF AUTHOR CONTRIBUTIONS

Category 1

(a) Conception and Design: Marolf

(b) Acquisition of Data: von Stade, Marolf

(c) Analysis and Interpretation of Data: von Stade, Randall, Rao, Marolf

Category 2

(a) Drafting the Article: von Stade, Marolf

(b) Revising Article for Intellectual Content: von Stade, Randall, Rao, Marolf

Category 3

(a) Final Approval of the Completed Article: von Stade, Randall, Rao, Marolf

CONFLICT OF INTEREST

The authors have no conflicts of interest to declare.

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